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Full Legal Name: Aaron Nathanael

Institution: SMA KRISTEN PENABUR KOTA WISATA

Category: Senior

Class/Grade/Year: Class 12

Country: Indonesia

Submission Date: September 20, 2026

Registered Email Address: gamesoffull@gmail.com

Research Topic: Topology Optimization and Structural Validation of an Engine Mount Bracket Using Autodesk Fusion 360

Title: Topology Optimization and Structural Validation of an Engine Mount Bracket Using Autodesk Fusion 360

Research Problem

An engine mount bracket is designed to keep a vehicle's engine and its mount to the frame while supporting their heavy weight [1]. By this nature, it is entailed to carry the engine's weight and transfer forces caused by acceleration, braking, cornering, road impacts, and engine torque into the supporting structure. As a consequence to its responsibility, a bracket is regularly exposed to vibration, torque reactions, and pressure from repeated cyclic loading, and sometimes thermal effects [2]; which wear it off causing, in most cases, severe vibrations, loud noises, and potential damage to other parts, symptoms documented in vehicle-specific diagnostic guides [3] and general mechanic failure-diagnosis advisories [4]. For years, engineers have had only thought of utilizing conventional engine mount brackets, which are designed with a redundant amount of material in hopes of providing adequate strength and stiffness under engine weight and torque loading. While this approach is preferred for its simplicity, it may result in unnecessary mass, wasteful material usage, and increased manufacturing costs. Conversely, aggressive material removal can allow brackets to bend or compress beyond their safe or functional limits when given weight to resist upon. The research problem is the lack of a systematically optimized engine mount bracket that finds middle ground between using as little material as possible while still keeping an engine anchored. This study will investigate whether the software Autodesk Fusion 360 topology optimization can identify a more efficient material distribution and whether the resulting design remains safe after being remodeled and validated using finite-element analysis.

Existing Literature

Topology optimization is a method that determines what positions should contain less material than others within a defined design domain, aiming for efficiency in the distribution of material while satisfying performance requirements and manufacturing constraints [5]. For an engine mount bracket, topology optimization is used to reduce mass while maintaining acceptable strength, stiffness, displacement, safety factor, vibration behavior, and manufacturability — an approach that has already been demonstrated directly on engine mounting brackets using both standard density-based methods [6] and bi-directional evolutionary structural optimization (BESO) [7]. Engine mount brackets are important vehicle components that support the powertrain and transfer loads from the engine to the vehicle frame. During operation, they may be subjected to engine weight, torque reactions, acceleration and braking forces, and even road-induced impacts. These combined conditions require the bracket to have sufficient strength, stiffness, fatigue resistance, and durability, requirements consistent with finite-element findings on comparable bracket-mounting structures [8]. At the same time, reducing the bracket's mass is desirable because it can reduce material use, improve vehicle efficiency, and lower manufacturing cost, particularly when the optimized geometry is paired with additive manufacturing for production [9].

In Autodesk Fusion 360, the user defines the design space, preserved geometry, obstacle geometry, materials, loads, constraints, objectives, and manufacturing requirements. The software then generates design alternatives by removing material from regions that contribute less to the specified load cases and retaining material along important load paths. However, the accuracy of the result to industry needs depends strongly on the assumptions and conditions entered by the user. One of the main difficulties is representing the actual operating conditions of the engine mount bracket. The bracket does not normally experience only one static force; it is exposed to combined forces and moments that vary during acceleration, braking, cornering, engine operation, road impacts, and engine start-up or shutdown. If the applied loads or constraints in Fusion 360 do not accurately represent the real mounting system, the software may generate an inefficient or unsafe material distribution [10]. Simplifying bolted connections, rubber mounts, contact conditions, and frame stiffness can also affect the accuracy of the simulation [11].

Similarly, a March 2026 publication by Le et al. sought to validate a full topology-optimization (TO) workflow using ANSYS, based on the SIMP (Solid Isotropic Material with Penalization) method [12] to secure numerical stability and safety-driven design for an aircraft engine lifting bracket, wishing to prove that shape is trustworthy enough to actually manufacture. To do this, Le et al. first calibrate their solver against the GE Jet Engine Bracket, a titanium design space subjected to four distinct load cases, running a mesh convergence study to confirm that the resulting shape is not simply a mesh artifact. With the solver calibrated, they let the SIMP solve run, updating material density with each iteration until the solution converges. Because the raw density output is not manufacturable as it stands, the converged result is then converted into a smooth, solid CAD model. This model is pushed through two parallel checks which involve a FEA verification and a physical build that evaluates stress, safety factor, and deformation against the FKM Guideline, and a physical build in which the part is sliced, 3D printed, and post-processed to remove supports. Only a design that clears both checks counts as validated. To show the work is relevant outside of the aerospace industry, the same process is then applied to the Z-axis adapter on a Bambu Lab A1 Mini 3D printer, where two manufacturing variants (one additive, one CNC) are produced from the same optimized topology to weigh weight savings against production practicality. Where either the simulated or the physical result falls short of the design target, the setup is revised and the process repeats.

The results prove this out. On the GE Bracket benchmark, the objective function takes a show by iteration number 10, and the final topology generated design results in a 75.23% reduction in mass while holding a static safety factor of 1.43 under the FKM Guideline, a number that meets design requirements even though some localized stress concentrations show up in the model. The Z-axis adapter tells a similar story at consumer scale: the additive variant achieves a 15.38% mass reduction with a safety factor of 1.65, while the CNC variant cuts slightly less mass, 14.42%, but still lands a safety factor of 1.52. Every variant is FEA-verified, and, crucially, none of it stops at the simulation stage. Physical prototypes are actually built for each one, so the manufacturability claim rests on parts that exist, not just on numbers that check out on screen [12].

Research Question

This study seeks to answer the research question: How can topology optimization generated by Autodesk Fusion 360 improve the structural performance and manufacturability of an engine mount bracket by controlling maximum displacement, maximum stress, and safety factor? This question is clear, focused, and relevant because it examines whether a computer-generated optimized design can provide improved structural efficiency while remaining practical to manufacture.

Methodology

The method is a quantitative approach. It will measure and compare the structural and manufacturing performance of an original engine mount bracket and a topology-optimized and remodeled bracket.

1. Hardware and Software Setup

The three-dimensional engine mount bracket model will be developed and tested using Autodesk Fusion 360 version 2704.1.36 on a Windows 11 laptop equipped with an AMD Ryzen 7 4800H processor with Radeon Graphics at 2.90 GHz, an NVIDIA GeForce GTX 1660 Ti graphics card with 6 GB of memory, and AMD Radeon graphics with 496 MB of memory. Fusion 360 will be used to create the CAD model, perform the structural study and Shape Optimization, and record the resulting measurements.

2. Baseline CAD and Structural Analysis

First, the original engine mount bracket will be opened in Fusion 360 and imported to the Simulation workspace. The material, geometry, units, and model dimensions will be checked before the analysis. Structural constraints will then be applied to the actual mounting regions, while forces representing the selected engine and operating loads will be applied to the appropriate interfaces. Critical regions, such as bolt holes, bolt sleeves, mating faces, and engine-mount interfaces, will be defined as Preserve Regions so they are not removed during optimization.

The original bracket will first be analyzed without optimization to establish baseline values for mass, maximum displacement, relevant maximum stress, and safety factor. The maximum displacement will be recorded in millimeters, while stress will be recorded in megapascals. The minimum safety factor reported by Fusion 360 will also be recorded. The same material, loading conditions, constraints, and analysis settings will later be used for the optimized and remodeled bracket so that the comparison remains consistent.

3. Shape Optimization

The Shape Optimization study will then be solved using Fusion 360. The software will calculate a new material distribution by removing material from regions that contribute less to the selected loading conditions while retaining material along the main structural load paths. The optimized result will be reviewed against the original model to ensure that all mounting interfaces, bolt locations, clearances, and functional surfaces remain available.

4. CAD Remodeling of the Optimized Result

The optimized result will not automatically be used as the final engineering model. Fusion 360 will superimpose the optimized organic mesh over the original bracket, allowing the design to be interpreted and remodeled. The optimized geometry will be used as a guide while standard CAD tools, including Sketch, Extrude Cut, Fillet, and other solid-modeling commands, are used to remove unnecessary material and create a clean, manufacturable bracket.

5. Structural Testing and Measurements

The remodeled optimized bracket will be returned to the Simulation workspace and analyzed using the same loading and constraint conditions as the original bracket. The following measurements will be collected:

- Mass, measured in kilograms.
- Maximum displacement, measured in millimeters.
- Relevant maximum von Mises stress, measured in megapascals.
- Minimum yield-based safety factor.

6. Measurement Standards and Acceptance Criteria

The maximum displacement will be considered acceptable when it remains below the allowable displacement established from the bracket's clearance, alignment, or design requirements. The maximum relevant stress will be considered acceptable when it remains below the allowable stress of the material.

For the preliminary static assessment, the safety factor will be calculated using the following equation [13]:

$SF = \sigma_{yield} / \sigma_{vM}$

where:

- SF = safety factor (dimensionless)
- σyield = yield strength of the bracket material, in megapascals (MPa)
- σvM = maximum von Mises equivalent stress obtained from the finite-element simulation, in megapascals (MPa)

A minimum safety-factor target, such as 2.0, will be selected and justified in the study. The optimized bracket will be considered acceptable for the preliminary static assessment when its minimum relevant safety factor is equal to or greater than the selected target. The safety factor will be evaluated together with maximum displacement and stress, rather than being used as the only measure of safety.

At least three repeated runs or result checks will be performed where practical. The mean value and variation will be recorded. The coefficient of variation may be calculated using the following equation:

$CV = (\sigma / \mu) \times 100\%$ [14]

where:

- CV = coefficient of variation, expressed as a percentage (%)
- σ = standard deviation of the repeated measurement results
- μ = mean (average) of the repeated measurement results

A low coefficient of variation indicates that the result is consistent under the selected conditions. Measurement uncertainty will also be discussed because simulation results depend on mesh quality, idealized constraints, material data, and load assumptions.

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